Atis Manustrakan

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github.com/atismanustrakan | atis-portfolio-eta.vercel.app

Education

University of Central Florida — Orlando, FL
Bachelor of Science in Computer Science — *Expected May 2029*

- **GPA:** 3.65/4.0 | **Honors:** Dean's List, Bright Futures Scholarship, Director's Award

Technical Skills

Languages: Python, C, Java, JavaScript, TypeScript
Frameworks & AI: React, Node.js, Express.js, Gemini API, MLC LLM, vLLM, ThunderAgent
Tools & Technologies: Git, GitHub, VS Code, CMake, Conda, Metal, REST APIs

Experience

Undergraduate Researcher — Secure & Fault-Tolerant LLMs — Fall 2026 – Present
University of Central Florida — *Orlando, FL*

- Collaborating with a research team to deploy and benchmark open-source LLM inference pipelines using MLC LLM, vLLM, and ThunderAgent on local hardware.
- Benchmarked 4-bit Qwen2.5-0.5B-Instruct on Apple Silicon with Metal GPU acceleration, achieving a decoding throughput of up to 107 tokens per second.
- Contributing to investigations of LLM runtime security, fault tolerance, memory corruption, bitflip faults, and performance optimization for embedded systems.

Undergraduate Researcher – Wearable Gait Monitoring System — Fall 2026 – Present
University of Central Florida — *Orlando, FL*

- Collaborating with an interdisciplinary research team to develop embedded software for a batterypowered wearable gaitmonitoring system.
- Implementing sensor communication and data acquisition logic to collect pressure sensor readings and IMU measurements.
- Supporting team evaluation efforts by processing gait data and assessing data quality, repeatability, and device performance.

Projects

GatorAnalyst | *TypeScript, React, Node.js, Gemini API*

- Collaborated with teammates to build a full-stack stock analysis platform using TypeScript, React, and Node.js to aggregate and visualize market data and company metrics.
- Integrated the Gemini API to simulate five distinct investor personas, each analyzing and rating stocks based on different investment strategies and risk preferences.

NutriPlanner | *Python, Gemini API, REST APIs*

- Collaborated with teammates to build an AI-powered nutrition planning application in Python using the Gemini API to generate meal recommendations based on user goals and dietary preferences.
- Worked with team members to integrate REST APIs for processing nutrition data, tracking macronutrient intake, and generating personalized recommendations from user inputs.

Organizations & Leadership

Knight Hacks — January 2026 – Present
Team Lead & Active Member — *Orlando, FL*

- Led a 3-person team to design and deploy a full-stack application, assigning development tasks and coordinating Git workflows..
- Collaborated with team members to integrate frontend and backend components, troubleshoot implementation issues, and manage development through Git.

Society of Asian Scientists and Engineers (SASE) — August 2025 – Present
Active Member — *Orlando, FL*